

Block light updates don't cross chunk borders properly when updated

MC-172980 Lighting

Status	Resolved / Fixed
Affected	20w08a, 20w12a, 20w15a, 20w17a, 20w19a, 1.16 Pre-release 2 (+51 more)
Fixed in	20w22a, 23w17a
Reported	2020-02-23

STEPS TO REPRODUCE

1. Create a new flatland world.
2. Set the time to midnight by running `/time set midnight`.
3. While standing on the ground, run `/fill ~ ~ ~ ~63 ~ ~63 sea_lantern`.
4. Enable the chunk border overlay with F3+G.
5. Observe that the light has not updated properly in some places.
6. Replace the sea lanterns with air by running `/fill ~ ~ ~ ~63 ~ ~63 air`.
7. Run `/fill ~ ~ ~ ~63 ~ ~63 sea_lantern` again.
8. Observe that the light again does not update properly in some places, particularly across chunk borders.

OBSERVED BEHAVIOUR

Actual behavior

Block light updates do not always propagate correctly across chunk borders after blocks are placed, removed, or replaced. When a light-emitting block such as a sea lantern is added or removed near or across a chunk boundary, some neighboring chunks retain stale block-light values instead of being recalculated.

This can produce visible lighting seams that align with chunk borders: areas with the same block arrangement may appear to have different brightness, and some blocks may remain incorrectly dark or illuminated. The affected light levels can sometimes change or flicker when nearby blocks are updated, but the lighting may become incorrect again after further block updates. Bulk operations such as `/fill`, structure or feature placement, and certain generated or explosion-placed light sources can trigger the issue.

EXPECTED BEHAVIOUR

Block light should update correctly and consistently across chunk borders whenever light-emitting or light-blocking blocks are placed, removed, or replaced. After the affected area is filled or cleared, all blocks should display their proper light levels without dark sections, unexpected patches, or flickering. Lighting should remain accurate regardless of whether changes result from commands, explosions, world generation, or ordinary block updates.

ENVIRONMENT

Minecraft Java Edition Versions: 20w08a, 20w12a, 20w15a, 20w17a, 20w19a, 1.16 Pre-release 2, 1.16.2 Release Candidate 2, 1.16.2, 1.16.3, 1.16.4 Release Candidate 1, 1.16.4, 20w45a, 20w48a, 20w51a, 21w03a, 21w05b, 21w06a, 21w19a, 1.17 Release Candidate 2, 1.17, 1.17.1 Pre-release 1, 1.17.1 Pre-release 3, 1.17.1 Release Candidate 1, 1.17.1 Release Candidate 2, 1.17.1, 21w40a, 21w41a, 21w42a, 21w43a, 21w44a, 1.18 Pre-release 1, 1.18 Pre-release 5, 1.18 Release Candidate 1, 1.18, 1.18.1, 1.18.2 Pre-release 1, 1.18.2, 22w12a, 22w13a, 22w14a, 22w16b, 22w17a, 22w18a, 22w19a, 1.19 Pre-release 3, 1.19, 1.19.1 Pre-release 4, 1.19.1 Pre-release 6, 1.19.1 Release Candidate 2, 1.19.1, 1.19.2, 22w43a, 1.19.3 Pre-release 2, 1.19.3, 23w03a, 1.19.4, 23w13a (includes snapshots)

ORIGINAL DESCRIPTION

The bug

Block light updates don't cross chunk borders properly when updated. This can be caused by fire placed by exploding beds, using `/fill` to fill a large amount of light sources or other blocks, using `/place` to place a feature or structure with light sources, or other miscellaneous circumstances. This may also occur upon world generation with

light sources, see

MC-249508

Resolved

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To reproduce

- Create a new flatland world

- /time set midnight

- Run

/fill ~ ~ ~ ~63 ~ ~63 sea_lantern

while on the ground

- Enable chunk border overlay (F3+G)

→ The light will not have updated properly in some places

- Replace the sea lanterns with air again

/fill ~ ~ ~ ~63 ~ ~63 sea_lantern

→ The light did not update properly again

Video

<https://youtu.be/lbIkC16GHYQ>

Original description

i was building a house and then all of a sudden i noticed that one half of the house was bright and lit while the other was mostly dark with light patches.

in the photo you can see the split. in the video i switch between the light block and dark block and they have the same light level even though they visually are different.

i had no lanterns before this and the same thing happened.

i tried updating the blocks around and it worked but then returned when i updated other areas.

you can see in the second video that it will just flicker when i try and update the blocks.

the other 2 screenshots are just other odd things that the lighting caused.

you should also know that i placed most of the roof with the /fill command if that helps

MANUAL RESULT

Version used	
Reproduced?	YES / NO / PARTIAL
Crash file	
Notes	
Tested by / date	

Live issue: <https://bugs.mojang.com/browse/MC/issues/MC-172980>